Shortcut-Resistant CAM Distillation for Long-Tailed Recognition

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Abstract

Real-world datasets often follow a long-tailed distribution, making generalization to tail classes difficult. We revisit this problem through the lens of shortcut learning, where models prefer the easiest predictive cues (e.g., background or textures) over object-centric semantics, especially under scarce and biased supervision. We find that this tendency is amplified for tail classes: limited examples often share similar contexts, making non-semantic signals highly correlated and thus tempting shortcuts, whereas head classes with diverse appearances and environments encourage more stable object-focused representations. Motivated by this observation, we propose Shortcut-Resistant CAM Distillation (SRCD), a plug-and-play framework that transfers object-focused explanations from head to tail classes. SRCD operates in the Class Activation Map (CAM) space, where a CAM provides a class-specific spatial evidence map for a prediction. SRCD aggregates CAMs from a small set of head-class candidates into a shortcut-resistant teacher using an energy-model weighting based on coherence and concentration, and distills it to the tail-class CAM. We provide a theoretical analysis that quantifies shortcut reliance as shortcut-region evidence mass in CAM space and shows that SRCD suppresses tail shortcuts. Extensive experiments on long-tailed benchmarks consistently improve strong baselines. The code is available at `https://github.com/Haifeng3/SRCD`.

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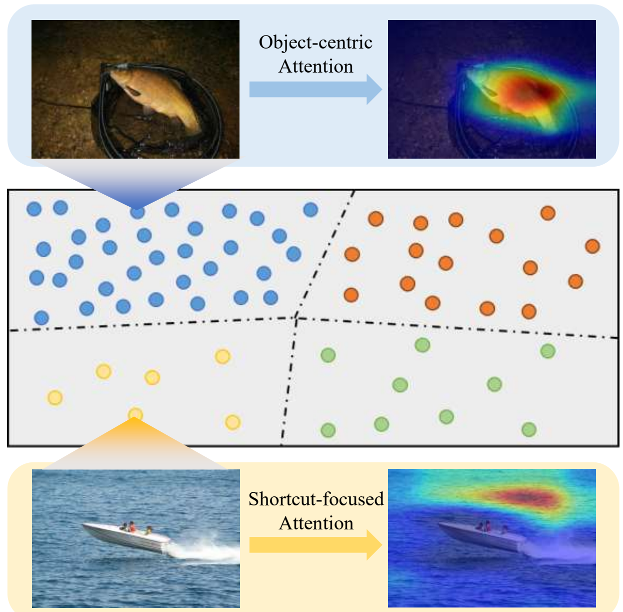

Figure 1. Shortcut learning in long-tailed recognition visualized by CAMs. For a head-class example (*Tench*), the model attends to the object region, while for a tail-class example (*Speedboat*), attention drifts to spurious cues (e.g., ocean background), leading to shortcut-dominated predictions and degraded tail generalization.

1. Introduction

In recent years, deep neural networks have become the dominant paradigm for visual recognition, enabling strong performance across a wide range of tasks such as image classification (He et al., 2016), medical image analysis (Ronneberger et al., 2015), and object detection (Ren et al., 2015). However, much of this progress implicitly relies on large-scale and relatively balanced training data, which rarely matches the distributions encountered in real-world deployments. In practice, data often follows a long-tailed distribution (Zhang et al., 2023), where a few frequent categories dominate the dataset while most categories contain only a limited number of examples. Learning effectively under such imbalanced supervision has therefore become a central challenge in modern computer vision (Zhang et al., 2025).

Despite steady progress (Zhang et al., 2025), tail-class generalization remains brittle: gains on tail categories are often